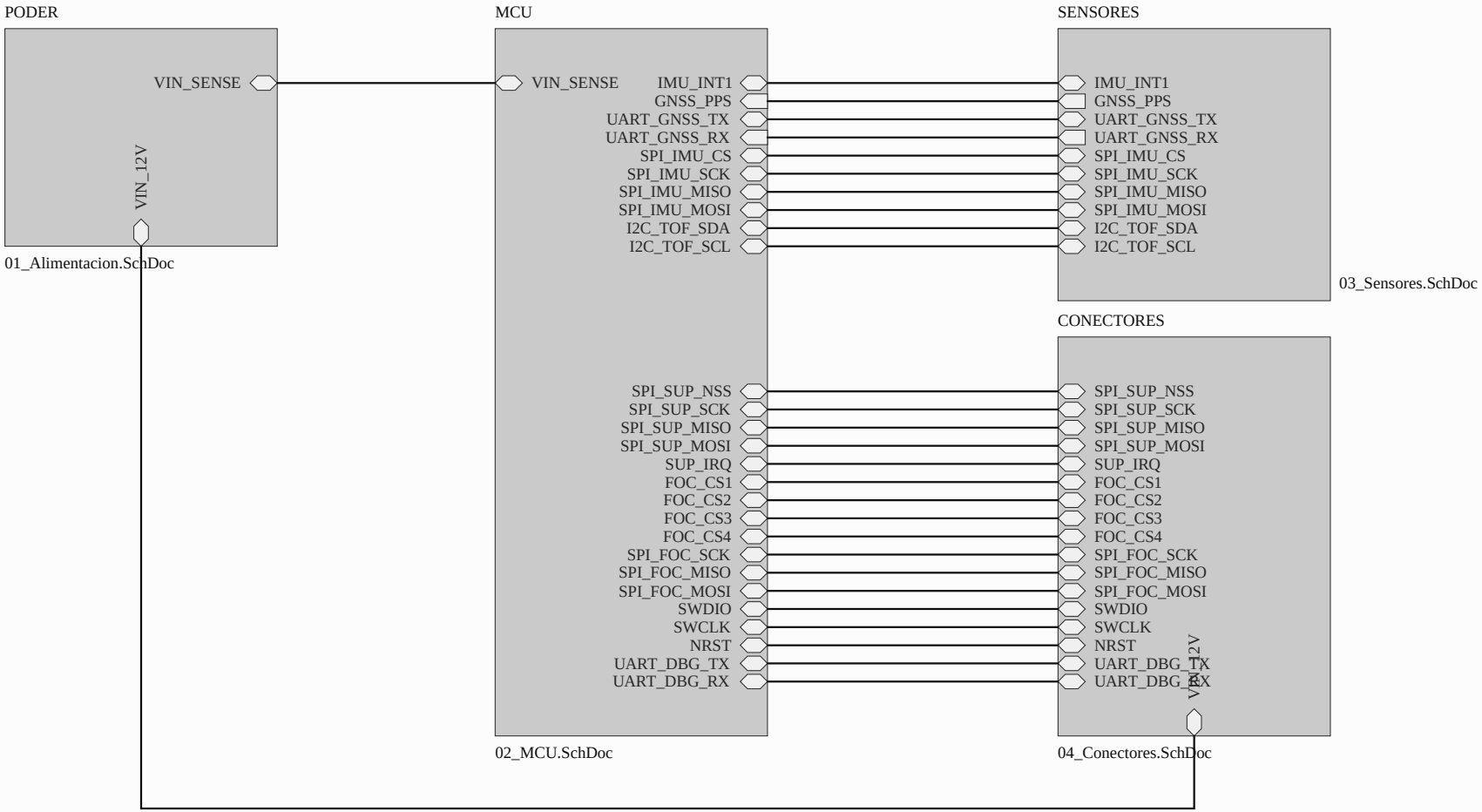


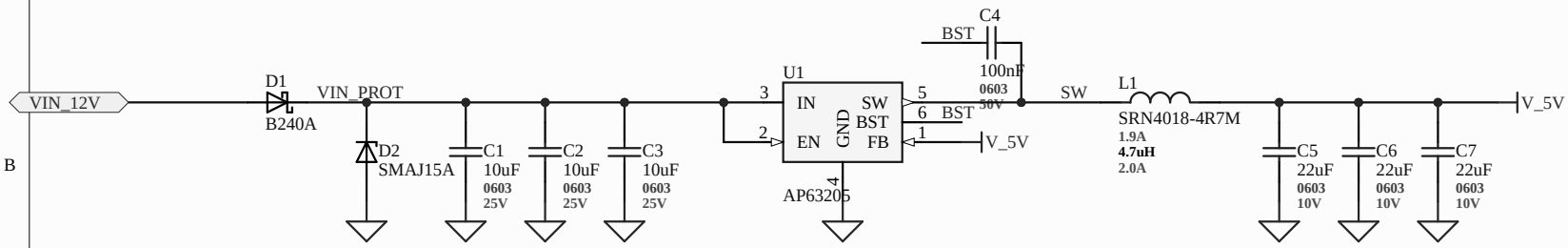
CONECTORES: SPI de entrada desde el supervisor (J1), SWD (J1/J2), UART de depuracion (J9) y bus SPI con 4 CS hacia los FOC (J3-J6)

Title	Modulo de Control Secundario: diagrama de modulos (hoja superior)	
Size: A	Number: ELO301-G1-00	Revision: 1.0
Date: 11/08/2026	Sheet 1 of 5	
File: C:\Users\..\00_Modulos.SchDoc	Drawn By: Grupo 1	

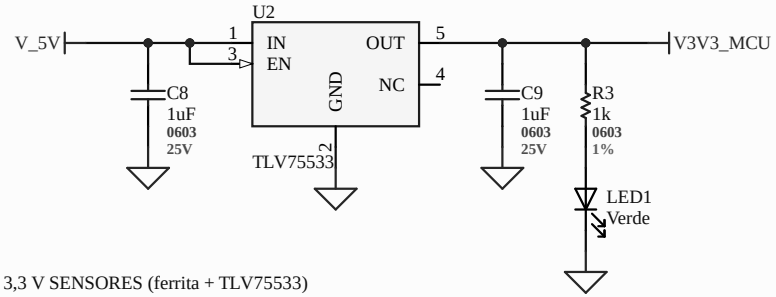
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BUCK 12 V -> 5 V: AP63205 a 1,1 MHz, L = 4,7 uH

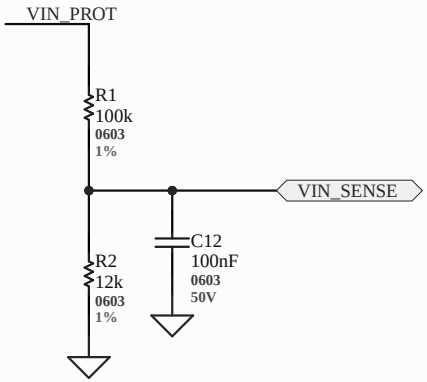
ENTRADA 12 V (conector de expansion): B240A serie + TVS SMAJ15A



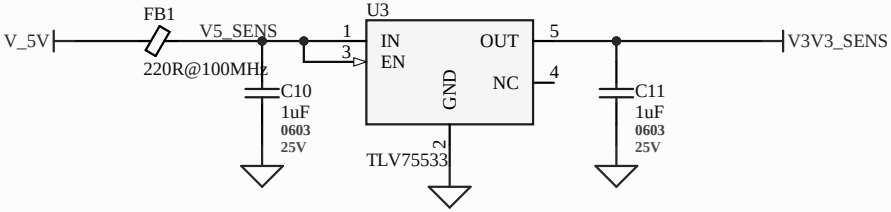
LDO 3,3 V MCU (TLV75533) + LED de encendido



MONITOREO VIN -> PA1 (ADC1_IN2): 100k / 12k + 100 nF

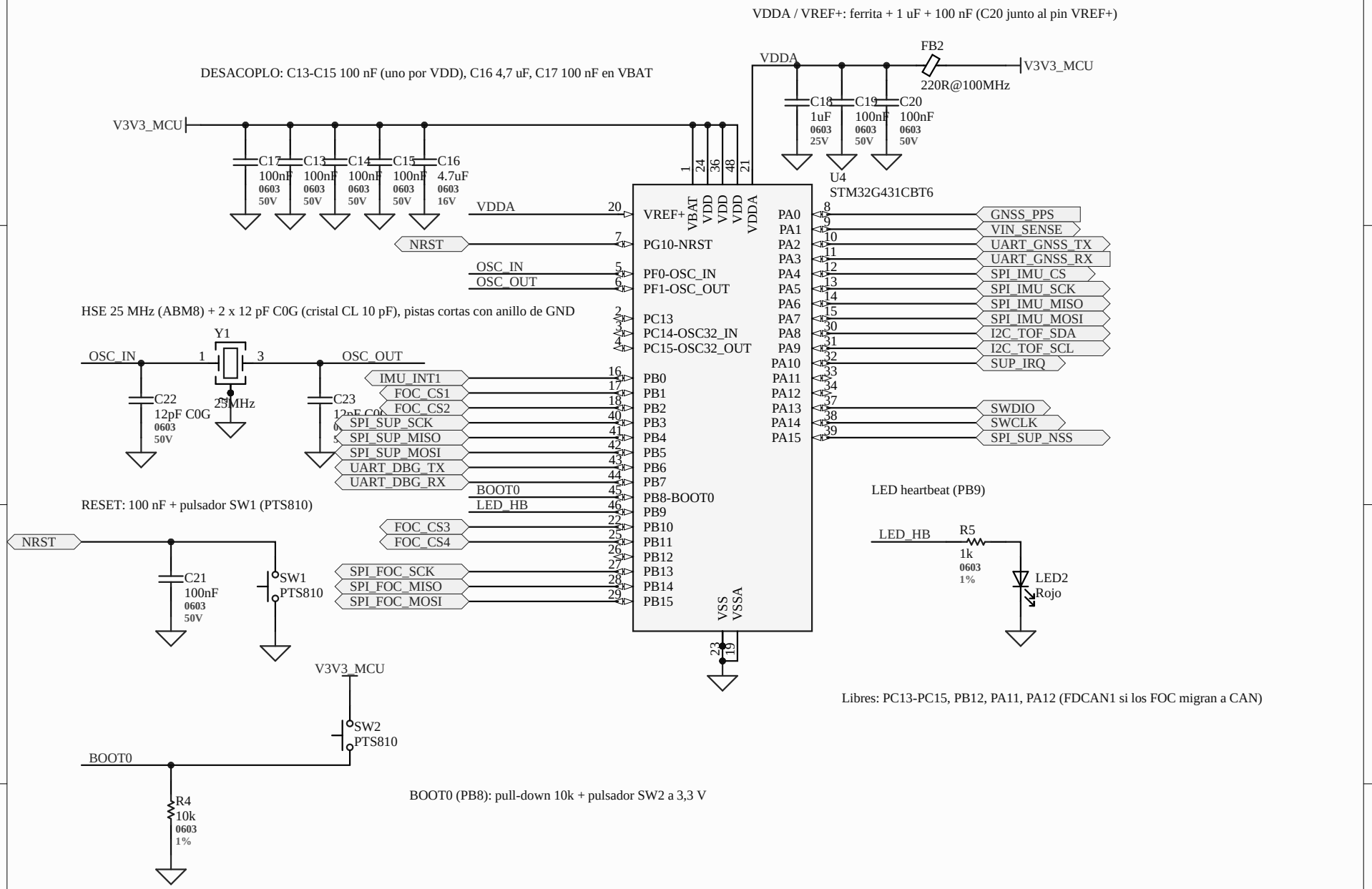


LDO 3,3 V SENSORES (ferrita + TLV75533)



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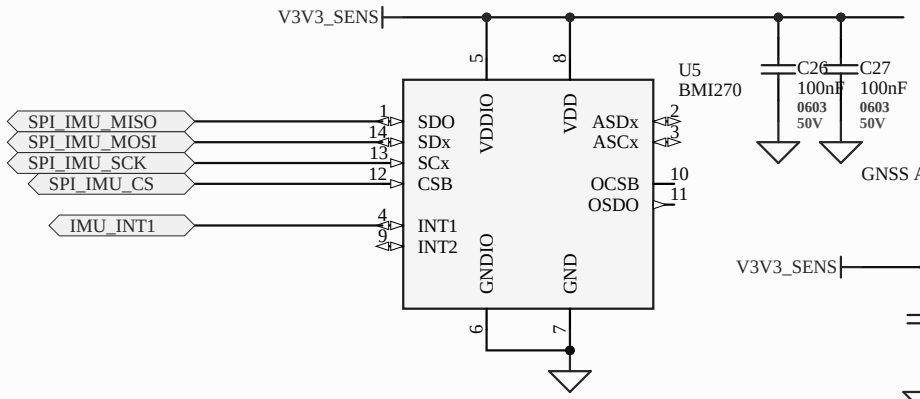
Title: Alimentacion: entrada 12 V, buck 5 V, LDO 3,3 V		
Size: A	Number: ELO301-G1-01	Revision: 1.0
Date: 11/08/2026		Sheet 2 of 5
File: C:\Users\..\01_Alimentacion.SchDoc		Drawn By: Grupo 1



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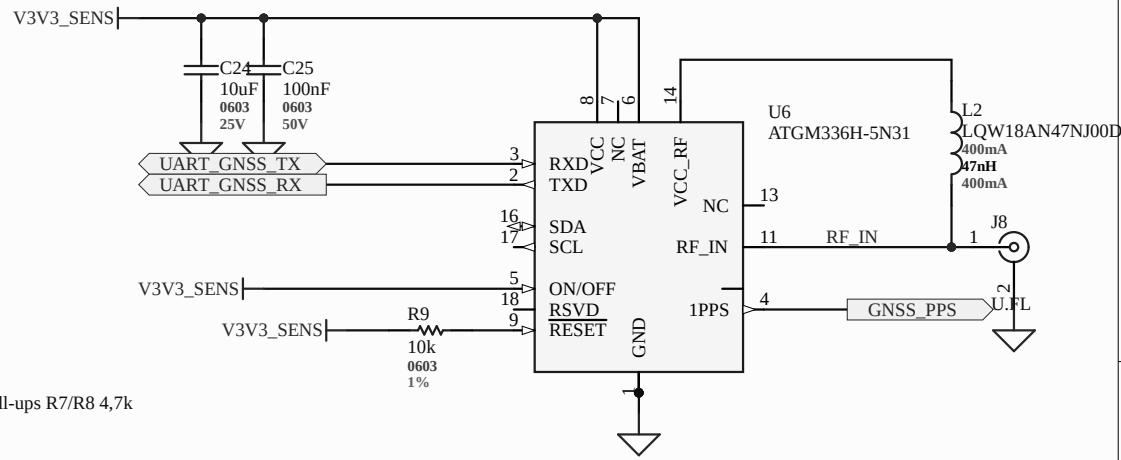
Title: MCU STM32G431CBT6		
Size: A	Number: ELO301-G1-02	Revision: 1.0
Date: 11/08/2026		Sheet 3 of 5
File: C:\Users\..\02_MCU.SchDoc		Drawn By: Grupo 1

IMU BMI270 por SPI1 (4 hilos): ASDx, ASCx, INT2, OCSB y OSDO sin conectar (Bosch, tabla 22)

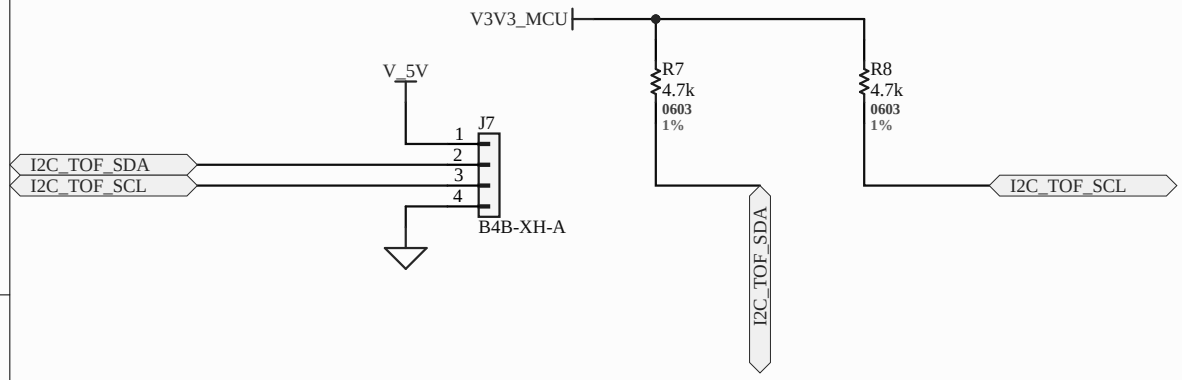


GNSS ATGM336H-5N31 (USART2 + PPS a TIM2). Antena activa por U.FL, linea de 50 ohm

VCC_RF -> L2 47 nH -> RF_IN (aplicacion ZKW)



ToF TFmini-S (externo) por I2C2: JST XH 4P, adaptador a GH en el arnes; pull-ups R7/R8 4,7k



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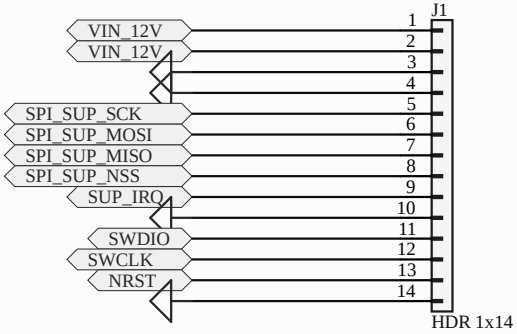
Title: Sensores: IMU BMI270, GNSS ATGM336H y ToF		
Size: A	Number: ELO301-G1-03	Revision: 1.0
Date: 11/08/2026	Sheet 4 of 5	
File: C:\Users\..\03_Sensores.SchDoc	Drawn By: Grupo 1	

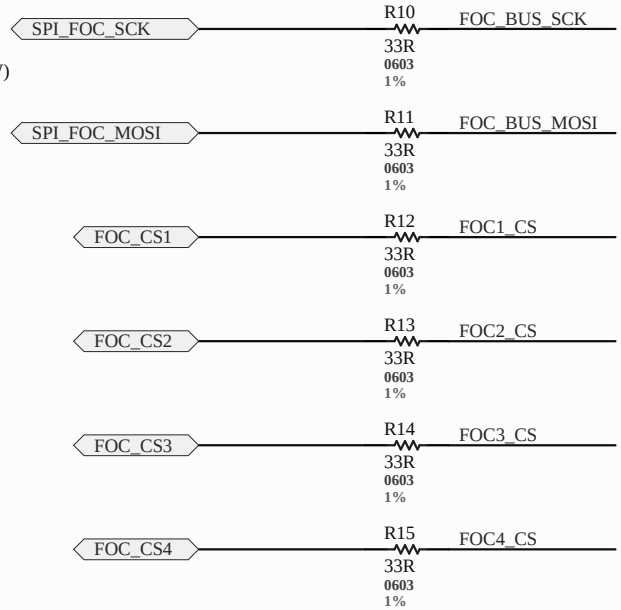
J3-J6: FOC 1-4 (JST XH 5P); PA11/PA12 libres para FDCAN1

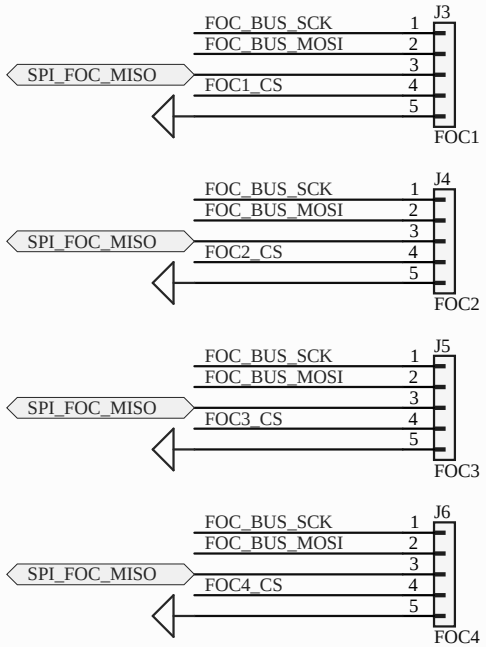
BUS SPI2 A LOS FOC: R serie 33 ohm en SCK, MOSI y CS (cable a 4 destinos); MISO directo

J1 EXPANSION (motherboard / Radxa Zero 3W)

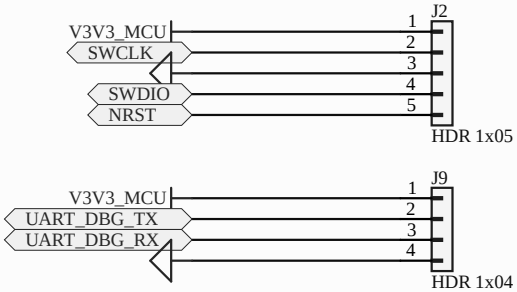
12 V, SPI3 esclavo + IRQ, SWD (Tabla 5)







J2 SWD (ST-LINK): 1 VDD, 2 SWCLK, 3 GND, 4 SWDIO, 5 NRST



H1-H4: perforaciones M3 con separadores (H3/H4 fijan el borde de los FOC)



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Title: Conectores: expansion, SWD y FOC		
Size: A	Number: ELO301-G1-04	Revision: 1.0
Date: 11/08/2026		Sheet 5 of 5
File: C:\Users\..\04_Conectores.SchDoc		Drawn By: Grupo 1